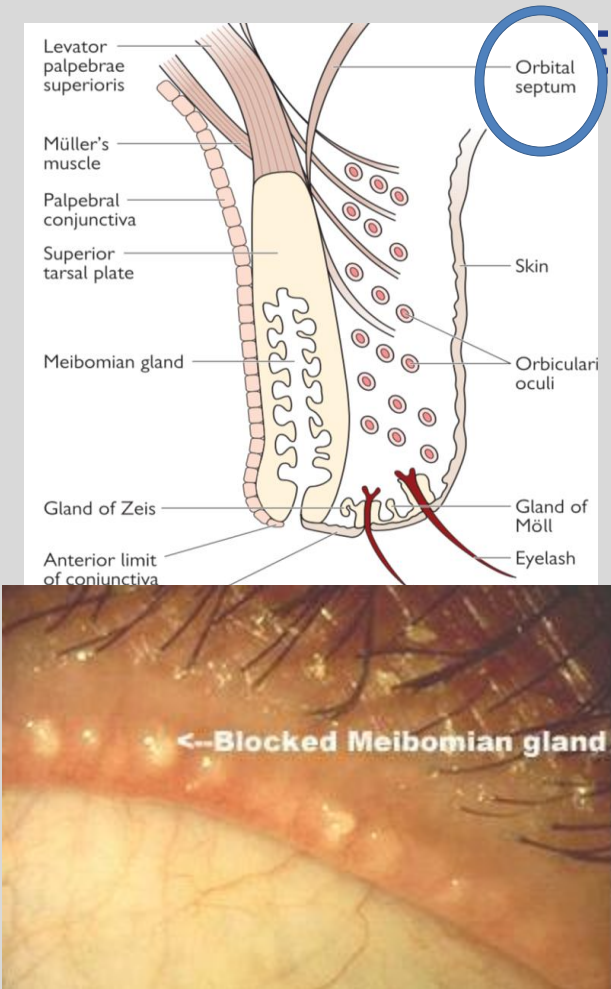


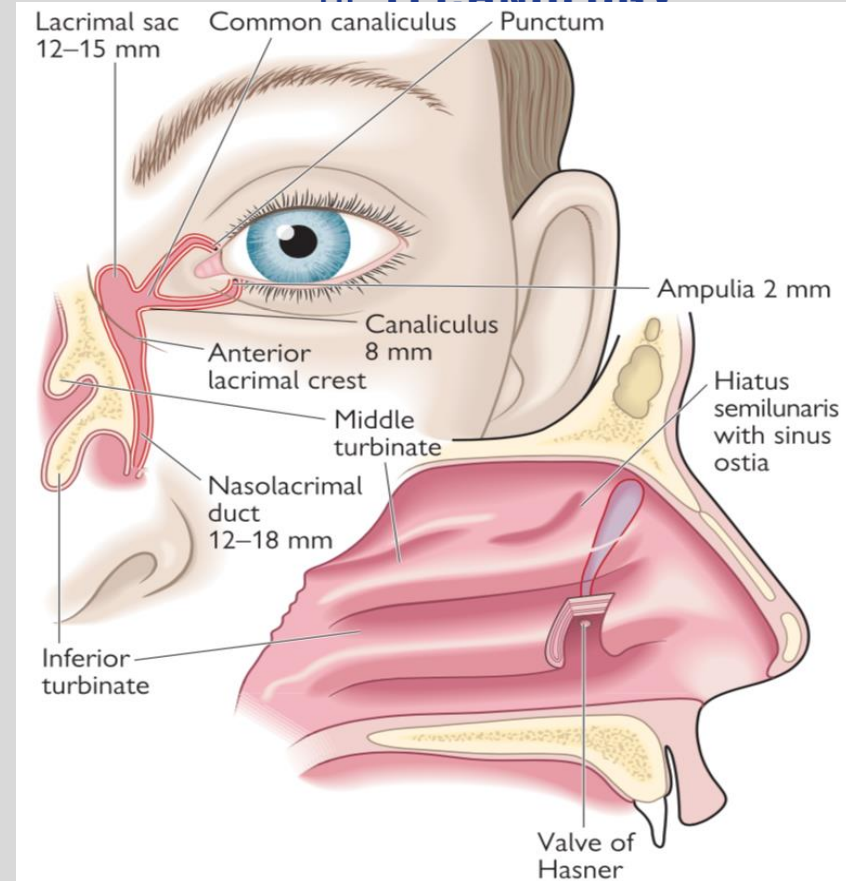
Ocular Anatomy: Eyelids

- Protect ocular surface from mechanical injury and when closed, protects the retina from bright light
 - Blinking assists distribution of tears across the ocular surface
 - Gross structure: lids meet at lateral and medial canthus
 - **Meibomian glands** (~50 per eye): produce sebaceous secretion that constitutes outer lipid layer of tear film
 - **Glands of Moll**: modified **sweat** glands that open on to the lid margin between the lashes or into a lash follicle
 - **Glands of Zeis**: **sebaceous** glands that open into each follicle



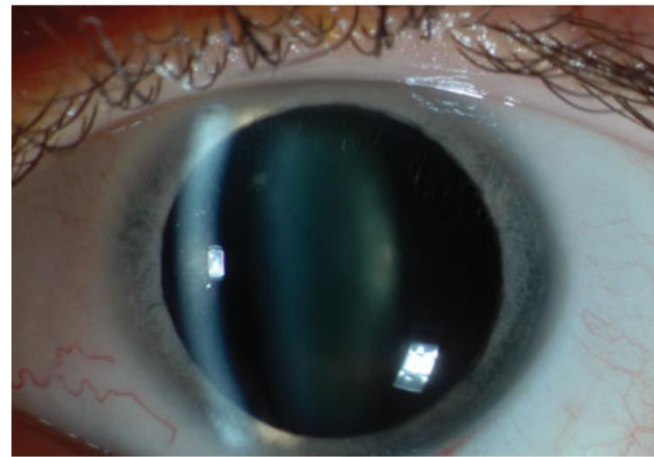
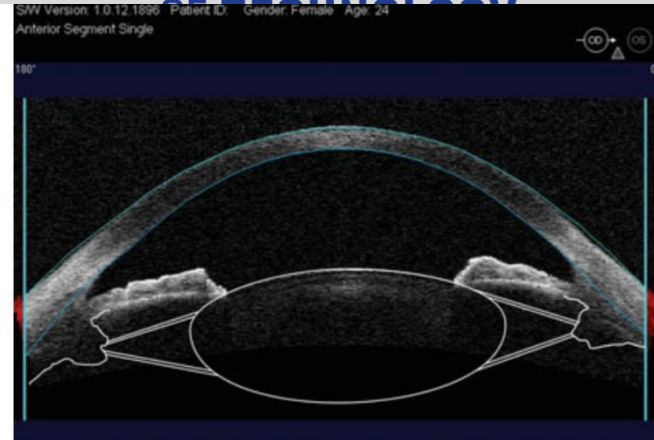
Ocular Anatomy: Nasolacrimal system

- **Tear production:**
 - main lacrimal gland
 - accessory lacrimal glands (in the conjunctiva)
- **Nasolacrimal drainage system:** conduit for the flow of tears from the external surface of the eye → nasal cavity
- **Lacrimal puncta:** small openings that lie on the medial aspect of the upper and lower lids
- **Lacrimal canaliculi:** begin at puncta superior and inferior and join together at common canaliculus to drain into nasolacrimal sac
- **Lacrimal sac:** 12mm long, within the lacrimal fossa in anteromedial portion of orbital wall
- **Nasolacrimal duct:** connects lower portion of lacrimal sac with the nasal cavity and opens into the inferior meatus of the nose



Ocular Anatomy: Lens

- Focuses images onto the retina, accounts for 1/3 refractive power of the eye
- **Zonules** support the lens within the eye: insert on lens equator circumferentially and are attached to ciliary body via ciliary processes
- **Accommodation**: ability of lens to change focus from distance to near; achieved by an increase in axial thickness of the lens and steepening of anterior and posterior radii of curvature
 - With age there is loss of accommodation: thus loss of clear near vision = need for reading glasses up close (*presbyopia*)



Ocular Anatomy: Extraocular muscles

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- move eyes in horizontal, *vertical* and **torsional** axes
- **Horizontal Muscles:**
 - **Medial Rectus:** adduction, Cranial nerve (CN)3
 - **Lateral Rectus:** abduction, **CN6**
- **Vertical Muscles:**
 - **Superior Rectus:** elevation, CN3
 - **Inferior Rectus:** depression, CN3
- **Oblique Muscles:**
 - **Superior Oblique:** intorsion, **CN4**
 - **Inferior Oblique:** extorsion, CN3
- **Synergists:** muscles in the *same* eye that move the eye in the same direction
- **Agonist-antagonist pairs:** muscles in *same eye* that have opposing actions
- **Yoke muscles:** pairs of muscles in opposite eyes that move the 2 eyes in the same direction
- **Strabismus:** improper alignment of the eyes



Basic Approach to “The Eye Patient”

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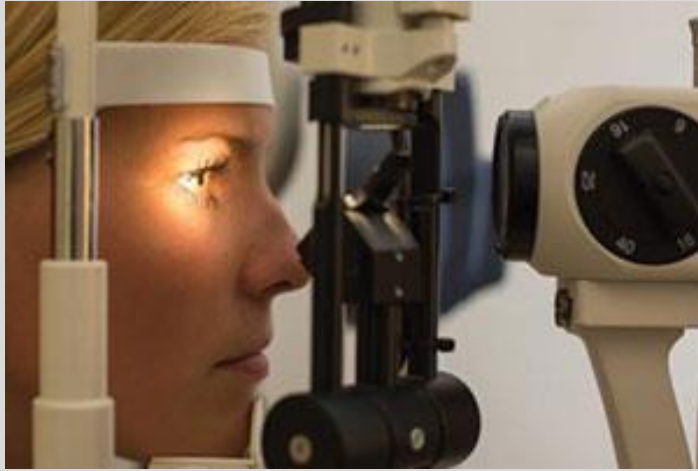
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- HPI: OPQRST, visual complaints, diplopia (monocular, binocular)
 - Recent hx of trauma? Surgery? Illness?
 - Exposure to someone with infectious eye condition (e.g. conjunctivitis)?
- Visual Acuity (VA, with and without correction)
- Extraocular motility (EOM)
- Pupils: size, reactivity, consensual response?
- Intraocular pressure (IOP)
- External exam: periocular skin changes? Edema?
- Slit lamp exam
 - Eyelids, lashes, conjunctiva, sclera cornea, anterior chamber, iris, lens, vitreous, retina, retinal vasculature
- Ancillary testing: as needed

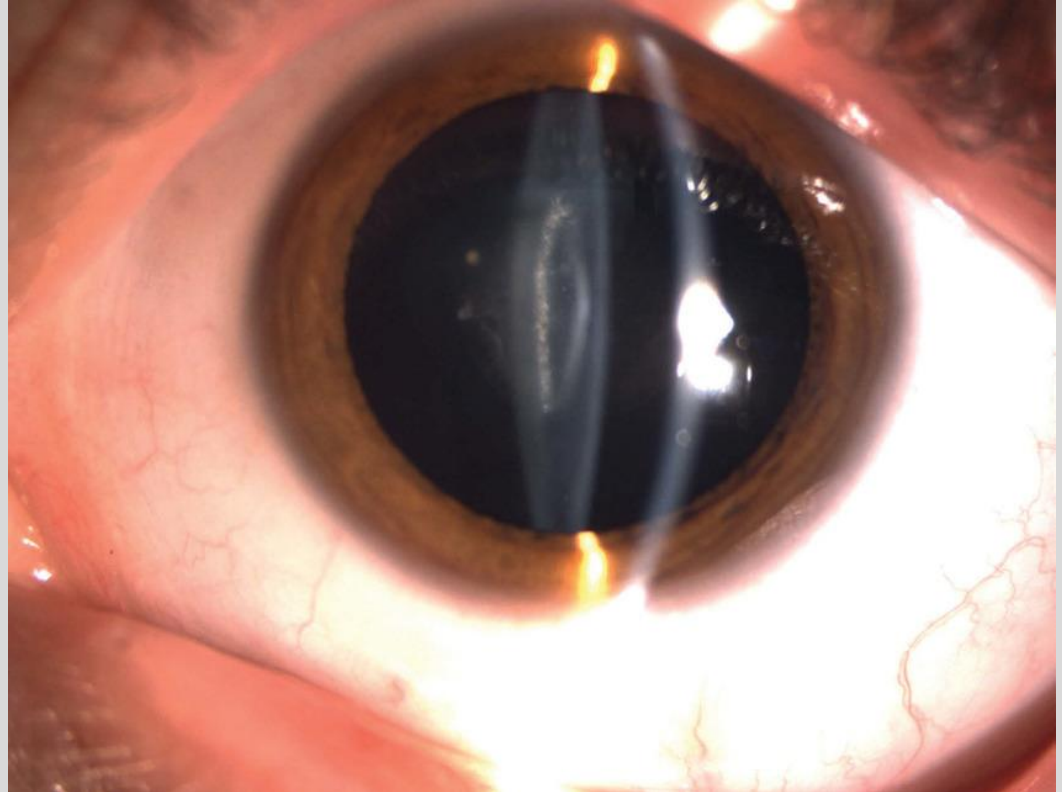
“The Eye Patient”

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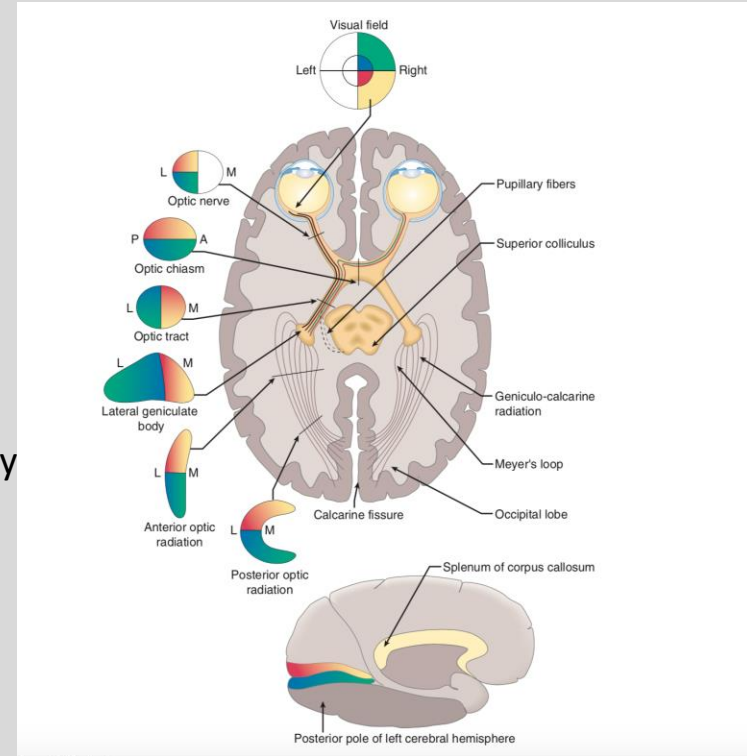
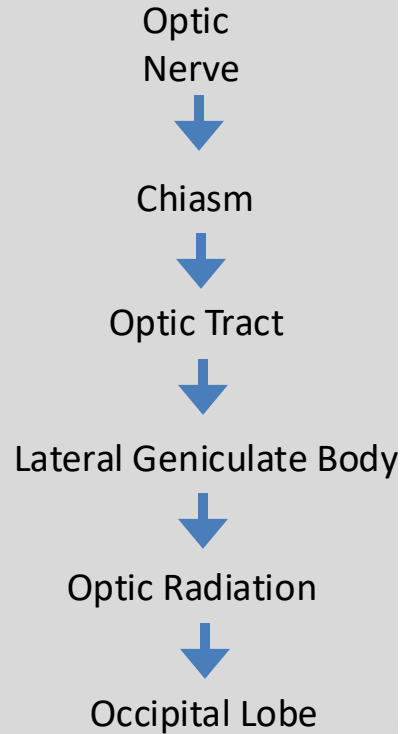
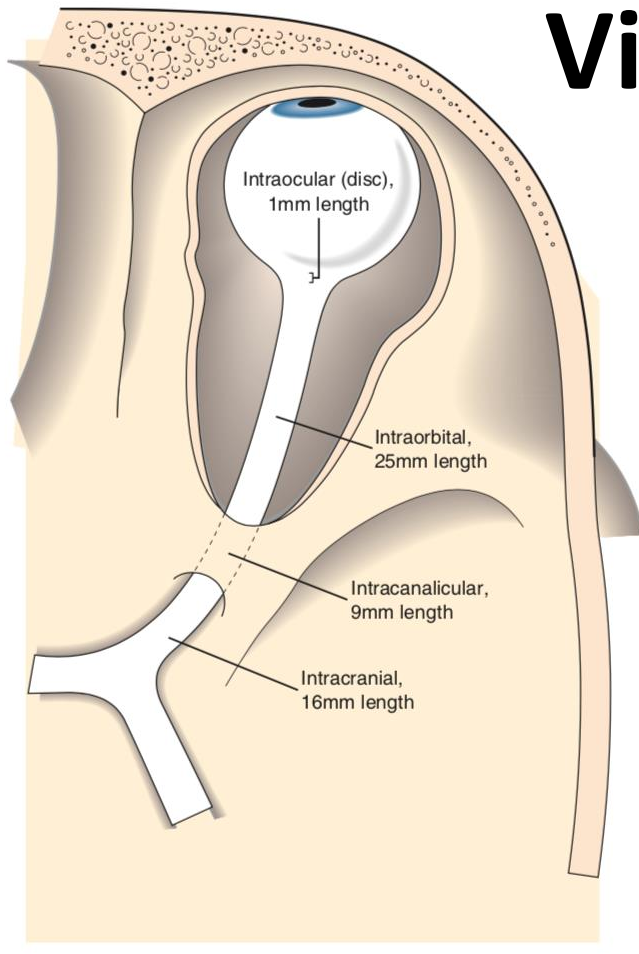


Slit Lamp



Visual Pathway

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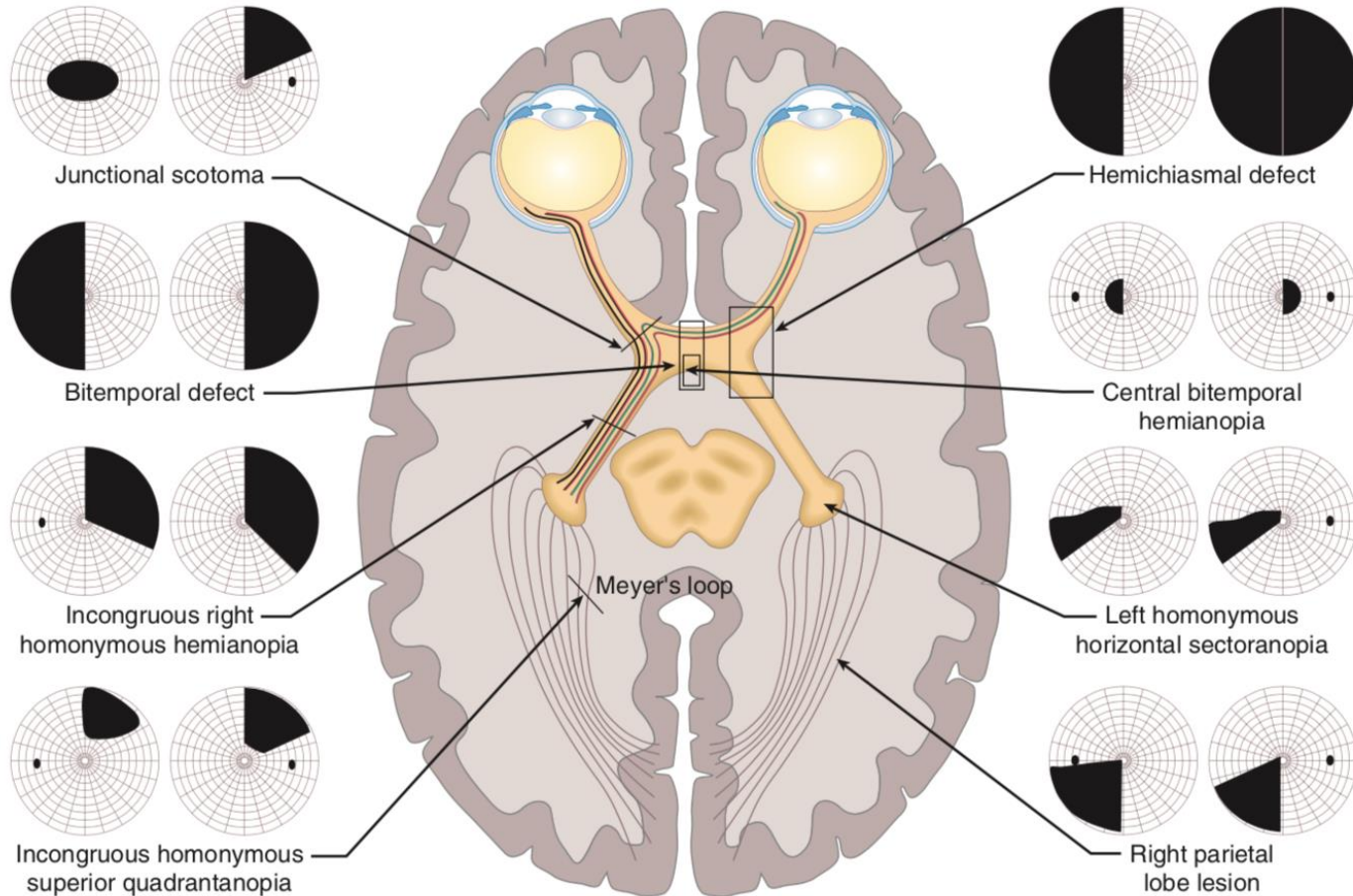


Visual Field Defects

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Pupils

- **Relative Afferent Pupillary Defect** (RAPD/Marcus-Gunn Pupil)
 - Light directed on one eye will *normally* induce consensual, constriction (miosis) in fellow eye. If there is an RAPD this does not occur (i.e. fellow eye will not react and/or dilate)
 - **Etiology:** large retinal lesion, asymmetric optic nerve disease (severe glaucoma), chiasm lesion, optic tract lesion (contralateral RAPD)
- **Light-Near Dissociation:** pupil does not react to light but near response is intact
 - **Etiology:** syphilis (Argyll-Robertson), Adie's pupil, familial dysautonomia (Riley Day syndrome), Parinaud's syndrome, encephalitis, alcoholism, herpes zoster ophthalmicus (HZO)
- **Anisocoria:** pupils of unequal size
 - **Miosis:** small pupil
 - Etiology: Horner's syndrome, pharmacologic (brimonidine, narcotics, pilocarpine, insecticides), Argyll-Robertson pupil, iritis, diabetes
 - **Mydriasis:** large pupil
 - Etiology: CN3 palsy, Adie's tonic pupil, pharmacologic (mydriatics, cycloplegics, cocaine), iris damage



Horner's Syndrome

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- “Ptosis, Miosis, Anhidrosis”
 - Droopy upper lid
 - Small pupil
 - Lack of sweating on forehead above affected eye
 - Etiology:
 - **Preganglionic:** from hypothalamus to superior cervical ganglion
 - Vertebrobasilar stroke, vertebral artery dissection, cervical disc disease
 - **Postganglionic:** superior cervical ganglion to iris dilator
 - **Internal Carotid Artery Dissection**: TIA, stroke, neck pain, amaurosis, dyacusia, dysguesia
 - Cavernous sinus disease, Herpes Zoster, tumor of parotid gland/nasopharynx/sinuses



More Pupils

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- **Hutchinson's Pupil:** unilateral dilated, poorly reactive pupil in comatose patient due to ipsilateral supratentorial mass → displacement of hippocampal gyrus (*uncal herniation*) entrapping CN3
 - Pupillomotor fibers travel in peripheral portion of nerve and are susceptible to early damage from compression



Ocular Muscle Disorders

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- **Chronic Progressive External Ophthalmoplegia** (CPEO)
 - Mitochondrial disease
 - *Kearns-Sayre Syndrome*: ophthalmoplegia with ptosis, retinal pigment degeneration and heart block
- **Myasthenia Gravis**: weakness of voluntary muscles due to autoimmunity to motor end plates
 - antibodies block acetylcholine (ACh) Receptors (binding, blocking, modulating)
 - Fatigability is hallmark
 - *90% have eye symptoms*
 - 50% with eye symptoms: eye fatigue, diplopia, limited EOM's
 - 80% with ocular presentation will develop generalized disease
 - Ocular myasthenia: ptosis and weakness of EOM only
 - Systemic myasthenia: other skeletal muscles involved
 - 5% of patients develop Grave's disease
 - Dx: **Tensilon test** (Edrophonium IV) ← Acetylcholinesterase (AChE) Inhibitor that prolongs action of ACh → stronger muscle contraction
 - Positive 80-90% of the time, high false negative rate
 - May cause severe bradycardia (thus need to give test doses of 1-2mg first and have atropine available for antidote if toxicity)
 - Rest test

Vascular Optic Neuropathies

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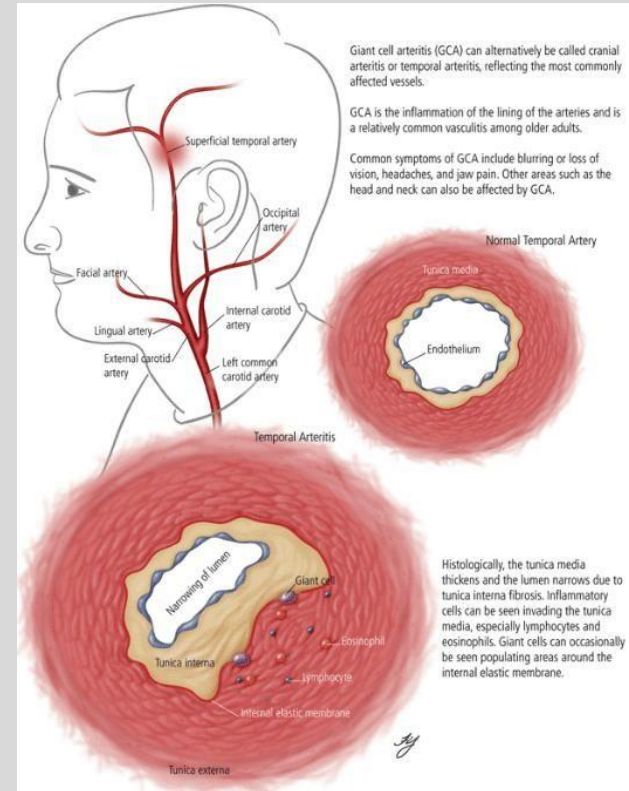
- **Anterior Ischemic Optic Neuropathy (AION):** infarct of optic nerve head just posterior to lamina cribrosa due to inadequate perfusion by posterior ciliary arteries → **acute painless vision loss**
 - Decreased color vision and visual acuity, RAPD, VF loss, unilateral optic disc edema, contralateral disc may appear small/crowded
 - **Arteritic ION (Giant Cell Arteritis/Temporal Arteritis):** inflammatory vasculopathy affecting medium-large size vessels
 - **Nonarteritic ION (NAION):** no associated sx, 50-75 y/o
 - Associated with DM, microvascular disease, collagen vascular disease, recurrence in same eye is rare, 25-40% fellow eye involvement, *NORMAL ESR*
- **Posterior ION:** acute, painless vision loss in one or both eyes
 - ↓ blood flow and oxygenation to the *intraorbital* optic nerve
 - ↓ in arterial perfusion pressure due to hypotension from volume blood loss
 - ↑ in peripheral vascular resistance causing reduction in downstream blood flow
 - ↑ in peripheral venous pressure from orbital edema
 - ↑ in intraocular pressure
 - ↓ in blood oxygen carrying capacity



Giant Cell Arteritis (Temporal Arteritis)

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- Inflammatory vasculopathy affecting medium-large size vessels
 - F>M (2:1), age>55 y/o
 - +/- amaraousis fugax/diplopia
 - **Systemic symptoms:** scalp tenderness, jaw claudication, joint pain (association with polymyalgia rheumatica), fever, malaise, anorexia, weight loss, anemia, Headache, tender temporal artery+/-decreased pulsation, neck pain, brain stem CVA
 - **Pathology:** granulomatous inflammation with epithelioid cells, lymphocytes, giant cells, disruption of internal elastic lamina, aneurysm formation
 - **Diagnosis:** elevated ESR, CRP, low hematocrit (HCT), thrombocytosis
 - **Treatment:** steroids (prednisone 60-100mg PO, +/- IV steroid initially) to **prevent fellow eye involvement** (65% risk of fellow eye involvement without treatment –usually affected within 10 days)



Idiopathic Intracranial Hypertension (*Pseudotumor*)

- **Papilledema with NORMAL neuroimaging and CSF**
- 90% female, mean age 33, associated with obesity
- **Findings:** HA, N/V, optic disc edema, transient visual obscurations, photopsias, VF defect, retrobulbar pain, pulsatile tinnitus, +/- vision loss, RAPD, diplopia, *elevated CSF pressure with normal composition (dx of exclusion)*
- **Etiology:** endocrinopathy, COPD, dural sinus thrombosis, use of steroids, OCP's, vitamin A, tetracycline, nalidixic acid use, isotretinoin
- **Tx:**
 - **No vision loss:** Headache (HA) treatment, weight loss
 - **Mild vision loss:** acetazolamide (Diamox), furosemide, weight loss
 - **Advanced vision/VF loss:** optic nerve sheath decompression, serial lumbar puncture (LP's), lumbar-peritoneal shunt



normal



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Optic Neuritis

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- Inflammation of optic nerve
- MC optic neuropathy in patients <45 y/o
- **Findings:** acute unilateral visual loss (may recover spontaneously)
 - ocular pain on eye movement
 - decreased color vision
 - Relative afferent pupillary defect (RAPD)
 - visual field (VF) defect
 - *optic nerve appears swollen or normal*
 - **Uhthoff's phenomenon:** worsening of symptoms with **heat** or **exercise**
 - **Phosphenes:** flashes of light induced by eye movements or sound
- DDx: idiopathic, MS, syphilis, sarcoidosis, Lyme disease, Wegener's granulomatosis, SLE, Devic's disease
- ***30% develop MS within 4 years; 38% develop MS within 10 years***



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Clinical Ophthalmology

Herpes Zoster Ophthalmicus

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- **Unilateral** painful skin rash in a **dermatomal distribution** of the **trigeminal nerve**
 - Vesicles typically crust and heal within 2-6 weeks
 - **Hutchinson's sign**: skin lesions at tip, side, or root of the nose—strong predictor of **ocular** involvement (nasociliary nerve)
- Typically occurs in older adults (>60 y/o); can present at any age and occurs after reactivation of latent varicella-zoster virus (VZV) present within the sensory spinal or cerebral ganglia
 - HIV patients 15x more likely to have disease
 - VZV: dsDNA virus in herpes family
 - Shingles vaccine now recommended for patients >60y/o
 - Many cases have a prodromal period of fever, malaise, HA, eye pain prior to skin rash eruption
 - +/- eye pressure, tearing, eye redness, decreased VA, pain in distribution of trigeminal nerve
- **Management**: careful eye exam
 - Oral (PO) antiviral (acyclovir, famciclovir, or valacyclovir); immunosuppressed patients/those unable to tolerate PO may require IV tx
 - Topical antibiotic ointment to prevent superinfection
 - Topical steroids for keratitis/uveitis
 - PO/IV steroids for scleritis, retinitis, choroiditis, optic neuritis
 - Amitriptyline 25mg PO for neuropathic pain
- *Prognosis: variable and dependent on long term sequelae*

Acute Angle Closure Glaucoma

- Major cause of blindness worldwide
 - Higher prevalence in Inuit and Asian populations
 - Associated with increased age and **hyperopia (Farsightedness)**
- Secondary to closure of the anterior chamber drainage angle
- Typically **acute, dramatic attack** with **increased intraocular pressure (IOP), nausea, vomiting, eye pain, eye redness, blurry vision, cloudy cornea, fixed, mid-dilated pupil**
 - “I was sitting in a dark room when I suddenly had eye pain”
- Management: topical antihypertensive drops: goal is to lower eye pressure!
 - In severe cases, IV acetazolamide/mannitol may be necessary
 - **True emergency** – elevated IOP for a long period of time may cause permanent vision loss!
 - +/- topical saline/glucose for cloudy cornea
 - **Laser peripheral iridotomy: definitive treatment**

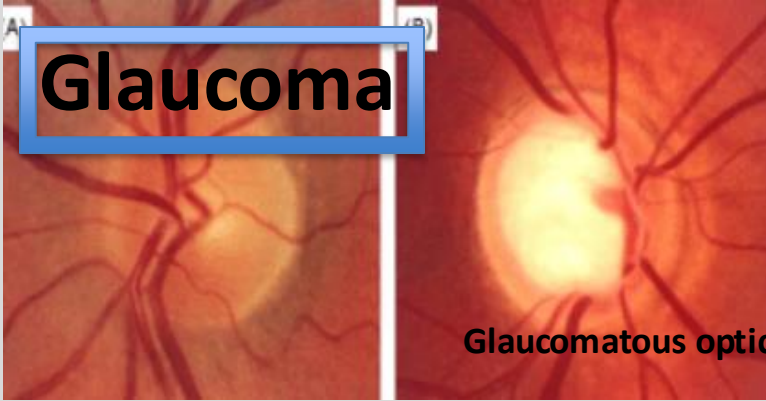
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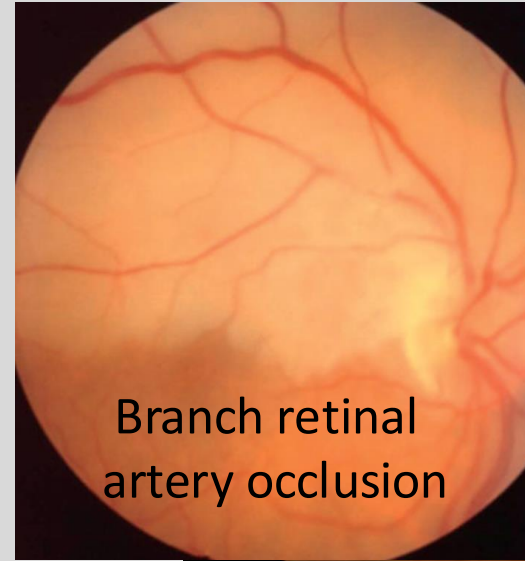
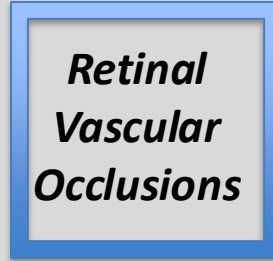
Decreased Vision

Glaucoma



Glaucomatous optic neuropathy

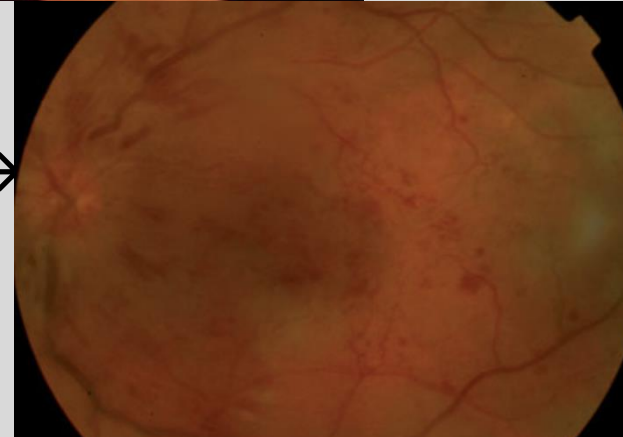
*Retinal
Vascular
Occlusions*



Branch retinal artery occlusion



Central retinal vein occlusion →

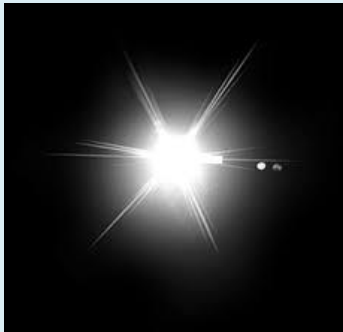


← Retinal Detachment

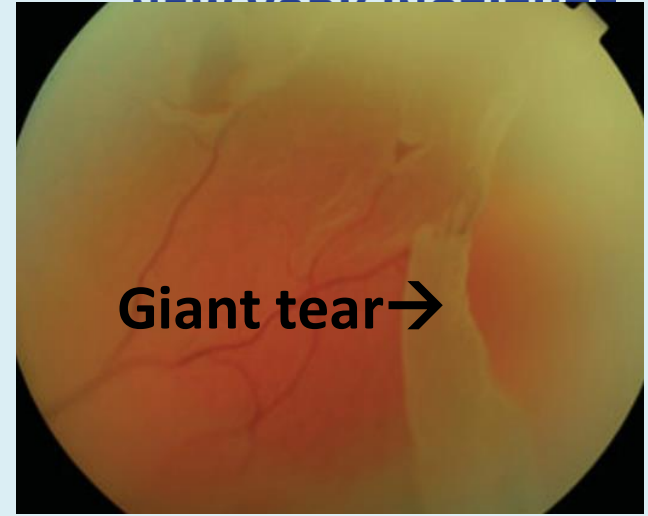
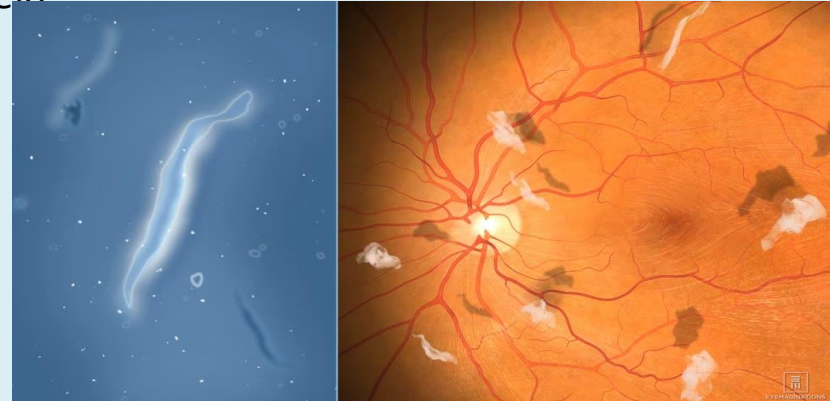
Retinal Detachment

- Patient complaint: **flashes/floaters**
 - **Pertinent ocular history:** high myopia, recent eye surgery/trauma, family hx of retinal detachment, severe diabetic retinopathy, vitreous traction, intraocular tumor
 - **Flash:** camera flash/lightning (not sparklers...)
 - **Floater:** multiple black spots
 - Curtain hanging down over/blocking vision/visual field
- Management: laser +/- vitrectomy surgery

“Flash”



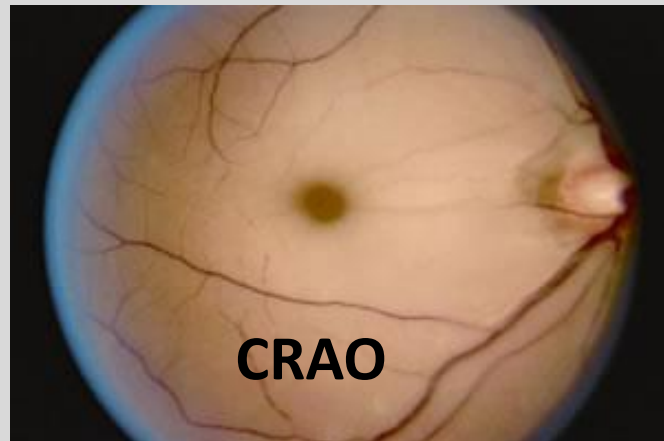
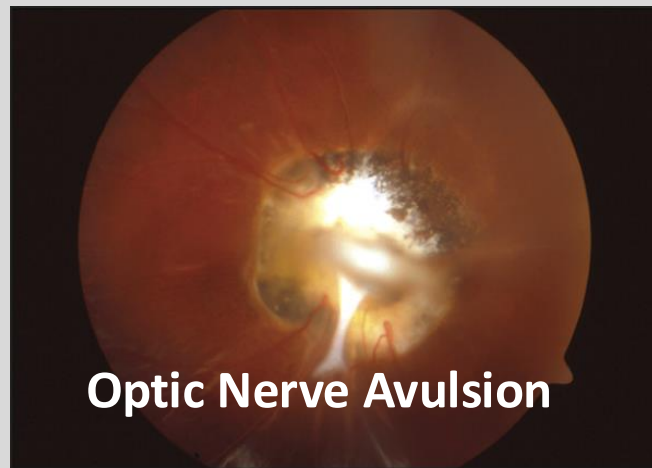
Floater



Vision Loss

- Traumatic Optic Neuropathy
- Anterior ischemic optic neuropathy
 - Arteritic (*inflammatory*): Temporal Arteritis
 - Non-arteritic
- Posterior ischemic optic neuropathy
- Central retinal artery occlusion (CRAO)
- Central retinal vein occlusion (CRVO)
- Vitreous Hemorrhage

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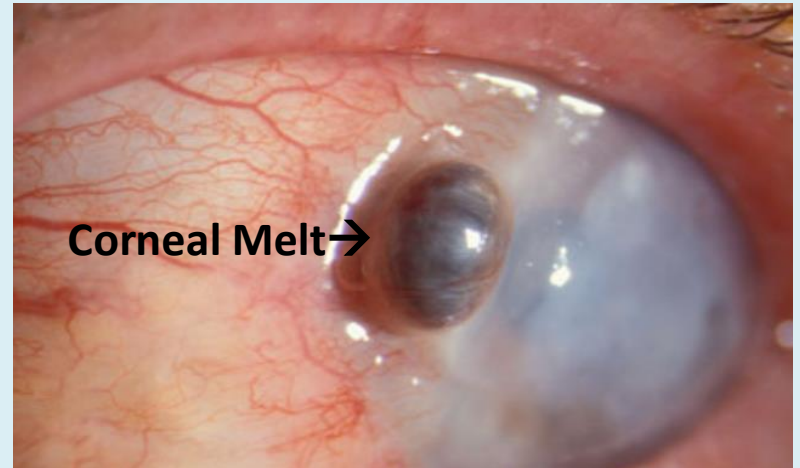


Corneal Abrasion

- Acute intense eye pain
- Usually h/o trauma or contact lens wear
 - *Recurrent erosion syndrome*: corneal epithelium irregular – easily abraded, therefore recurrent abrasions with minimal trauma (may even occur upon eye opening in the morning!)
- Management: antibiotic drops/ointment
 - +/- placement of bandage contact lens for large abrasions
 - Usually heals within a few days
 - DO NOT DISPENSE ANESTHETIC EYE DROPS→

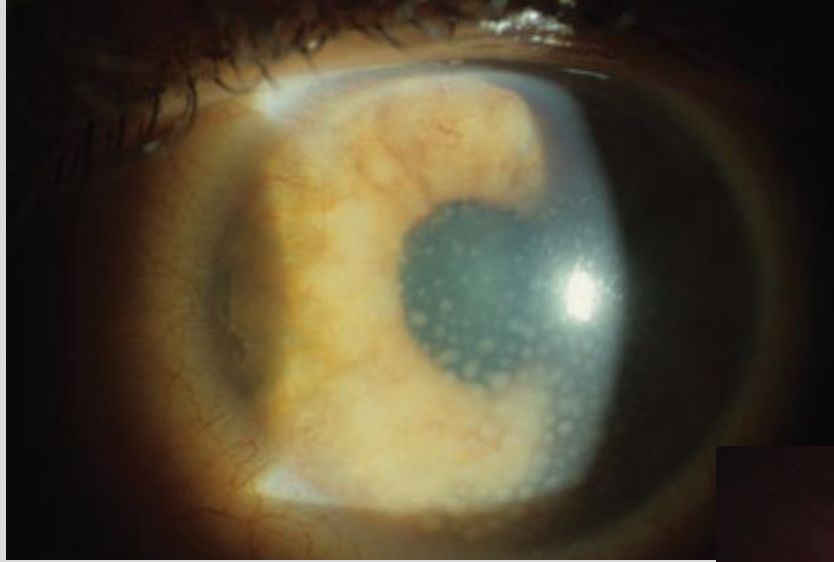


JTE



Chronic Eye Pain

- Dry eye
- Chronic Uveitis
- Migraine
- Sinusitis
- Scleritis
- Episcleritis

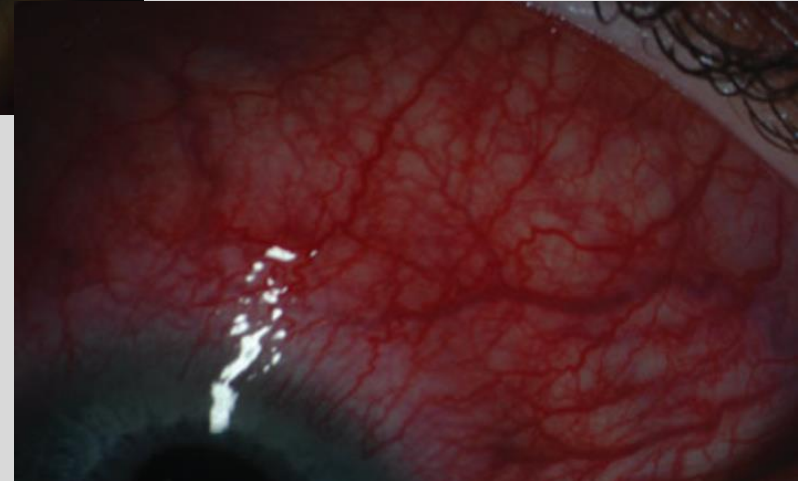


Diffuse Scleritis→

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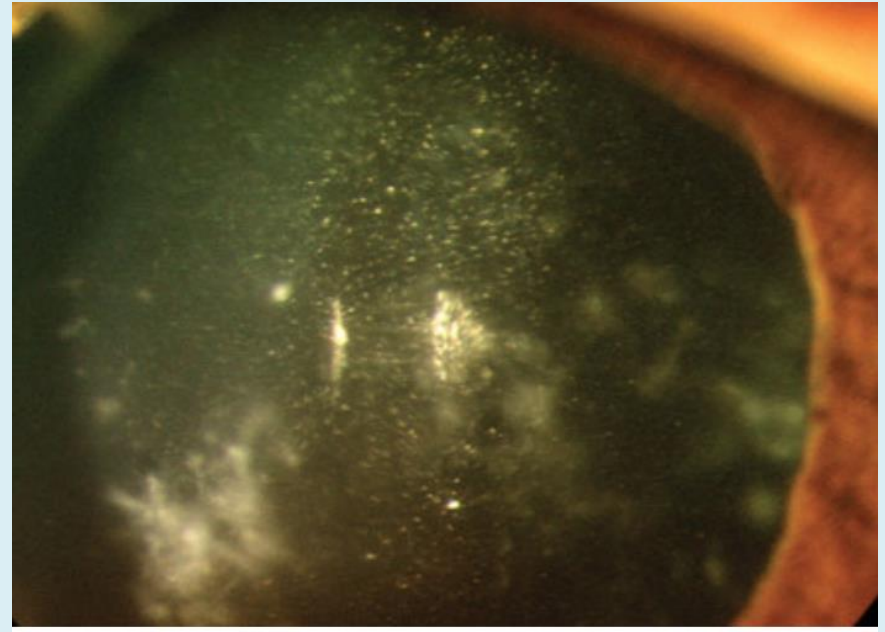
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**← Chronic Uveitis
with keratic
precipitates**



Uveitis

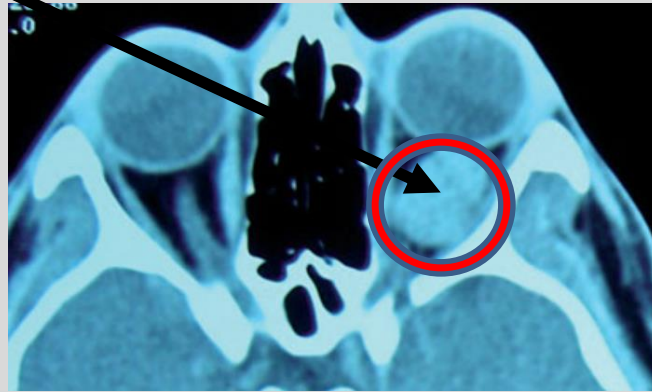
- Uvea: Iris, ciliary body, choroid
- Uveitis: inflammation of any/combination of these structures
 - Unilateral or Bilateral
- Etiology
 - Idiopathic – may only occur once!
 - Infectious
 - **Viral**: herpetic, HIV, CMV, TB, COVID-19
 - **Bacterial**: Syphilis, post-surgical infection
 - Autoimmune
 - Rheumatoid arthritis, lupus, Crohn's disease, Bechet's disease, Ankylosing spondylitis, Sarcoidosis
- Workup:
 - Patient history
 - Labs to check for infectious/systemic conditions that predispose to uveitis
 - Comprehensive ophthalmic exam
 - +/- referral to rheumatologist/infectious disease
- Treatment:
 - **Steroids**—topical, subconjunctival, periocular, intraocular, oral



- **Common symptoms**: dull achy pain, photophobia, blurry vision, floaters
- **Common signs**: poorly reactive pupil, conjunctival injection, keratic precipitates (KP's), white blood cells in the anterior (iritis) and/or posterior segment (vitritis)

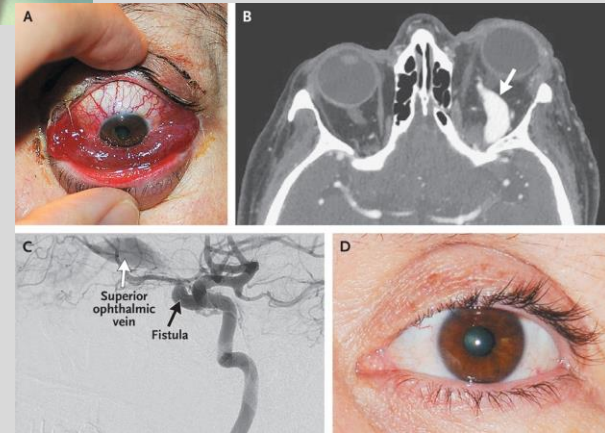
Diplopia

- Strabismus
- Thyroid Eye Disease
 - Proptosis
- Retroorbital Mass
- Ateriovenous malformation (AVM)
- Carotid-cavernous fistula



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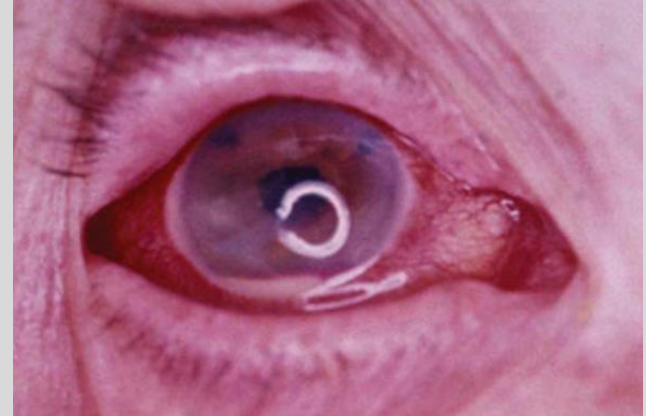


Carotid-cavernous fistula

Systemic Disease with Ocular Findings

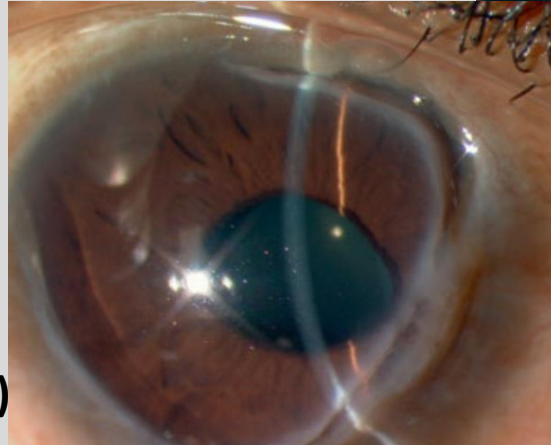
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- Rheumatoid arthritis
- Lupus
- Sjogrens syndrome
- Crohn's Disease/IBD
- Wilson's disease
- Juvenile rheumatoid arthritis (JRA)
- Hepatitis C
- Syphilis



Uveitis: autoimmune disease

**Corneal Thinning &
vascularization
(autoimmune disease)**



**Sunflower Cataract
(Wilson's Disease)**

Systemic Disease with Ocular Findings

- Sarcoidosis
- TB
- Sepsis
- HIV
- Diabetes
- Hypertension
- Leukemia
- Melanoma



Tuberculosis Retinitis→

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← HIV Retinopathy



Diabetic Retinopathy

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- Affects up of 80% of patient with Diabetes (I or II) >20yrs
 - Leading cause of blindness in patients 20-64 y/o
- Signs/Sx: blurry vision, distorted vision, floaters
 - Microaneurysms, retinal ischemia, macular edema, vitreous hemorrhage, retinal detachment (*a vasculopathy!*)
- Tx: GOAL IS TO reduce vision loss!
 - Laser surgery
 - Intravitreal injections: steroid, anti-Vascular endothelial growth factor (VEGF) agents
 - Vitrectomy surgery
 - GLYCEMIC CONTROL!

